Cache-Optimal Tiling for Mixed-Precision Matrix Multiplication

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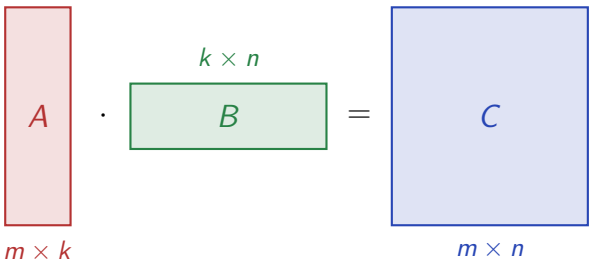
Matrix Multiply is Memory-Bound



for example:

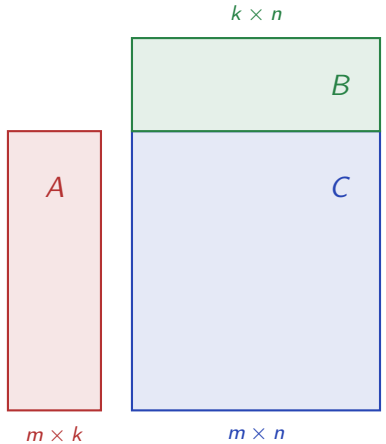
256×256 fp64: ≈0.5 MB

Typical L1 cache: 16–64 KB

⇒ 8×–32× **larger than L1**

Fix: *tile* the loop — bring in one smaller C tile and keep it resident in L1.

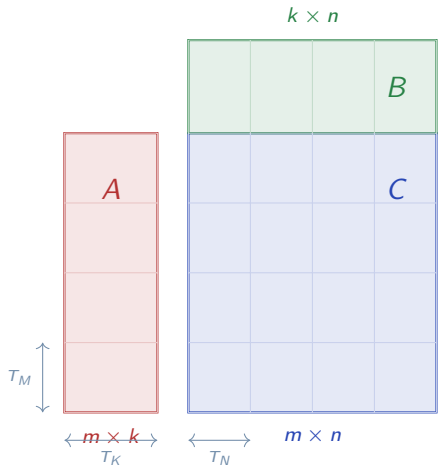
C-Stationary Tiling: Step by Step



$C = A \cdot B$

For large m, n, k — none fit in L1.

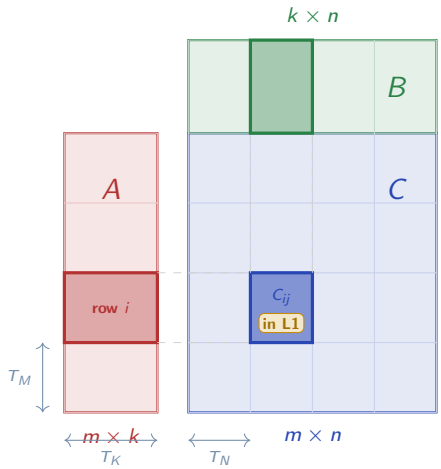
C-Stationary Tiling: Step by Step



Tile shape: $T_M \times T_N \times T_K$

Each matrix divided into a grid of tiles.

C-Stationary Tiling: Step by Step

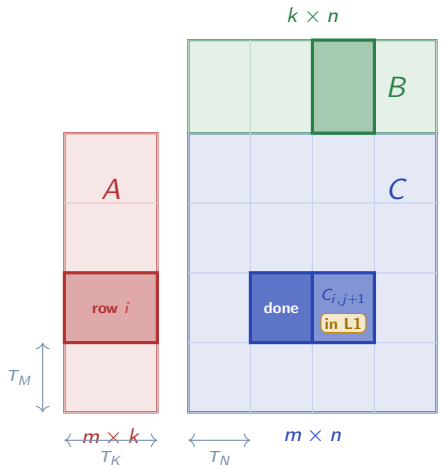


Fix C_{ij} in L1.

Stream A row-band and B col-band through it.

$C_{ij} \mathrel{+}= A_{\text{row } i} \times B_{\text{col } j}$

C-Stationary Tiling: Step by Step



C_{ij} done — move to B col $j{+}1$.

A row-band stays in L1.

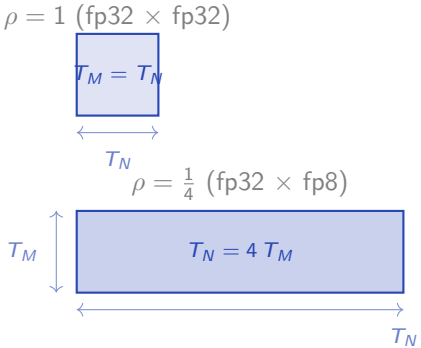
Asymmetric Precision $\Rightarrow$ Asymmetric Tiles

$$\#A\ Bytes = T_M \cdot k \cdot A_p$$
$$\#B\ Bytes = T_N \cdot k \cdot B_p$$
$$\Rightarrow T = \frac{MN}{T_M T_N}(T_M \cdot k \cdot A_p + T_N \cdot k \cdot B_p)$$

$$\boxed{T = MNK\left(\frac{A_P}{T_N} + \frac{B_P}{T_M}\right)}$$

Find best ratio:

$$\Rightarrow \quad \frac{T_N^*}{T_M^*} = \frac{A_P}{B_P} = \frac{1}{\rho}$$

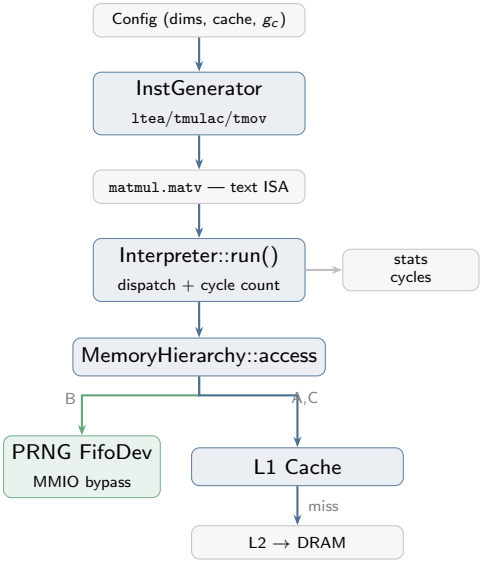


Same cache budget S — wider tile, better B reuse.

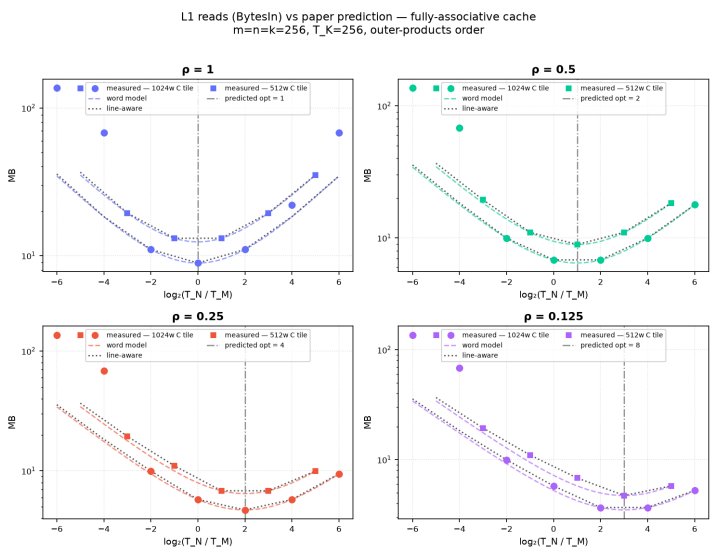
The Simulator We Built

Three main components:

- **InstGenerator** — compiles tile ops to a text ISA (`ltea/tmulac/tmov`)
- **Interpreter** — dispatches instructions, counts cycles per access
- **MemoryHierarchy** — configurable L1/L2/DRAM with per-access tracing



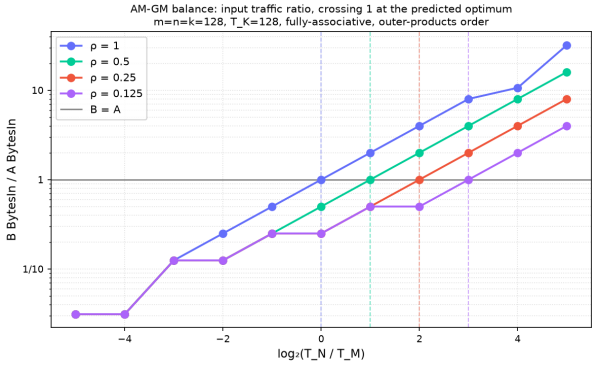
Traffic Minimum at the Predicted Tile



Each curve shows L1 reads as a function of tile aspect ratio T_N/T_M.

Dashed line: paper's prediction $T_N/T_M = 1/\rho$.

B/A Balance: Exact Confirmation



Each curve is the ratio of B reads to A reads as T_N/T_M varies.

Traffic saved vs. square tile:

ρ	saving	
1	0%	(symmetric)
0.5	0%	
0.25	18%	
0.125	36%	

Traffic $\neq$ Time

The paper minimizes *bytes transferred*:

$$T_{\text{traffic}} = MNK\left(\frac{A_P}{T_N} + \frac{B_P}{T_M}\right)$$

But cycles $\neq$ bytes.

We need a *cycle-accurate* model.

Enter the PRNG FIFO

Enter the PRNG FIFO

B elements are generated *on-chip* by a hardware random number generator, streamed directly into the compute unit.

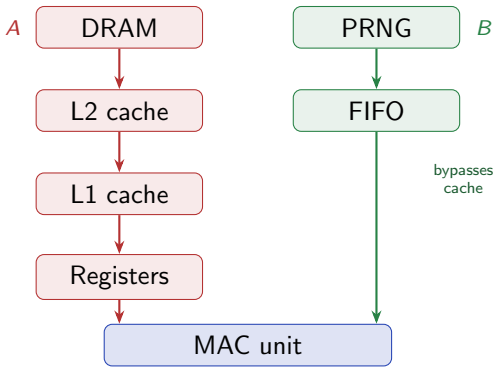
B never touches the cache.
B has no memory address.

The B_P / T_M traffic term **vanishes**.

Instead, B introduces a new cost: on-chip *generation time*.

$$\text{generation cost} = g_c \text{ cycles / element}$$

The PRNG FIFO: B Without Memory



Random matmul: $C = A \cdot B$ where B is a pseudo-random matrix generated on-chip.

Use cases: random projections, privacy-preserving ML, approximate algorithms.

Two independent bottlenecks

A-bound: load A from memory

B-bound: wait for PRNG (g_c cy/elem)

The PRNG FIFO: B Without Memory

$$\text{Runtime} = ?$$

The PRNG FIFO: B Without Memory

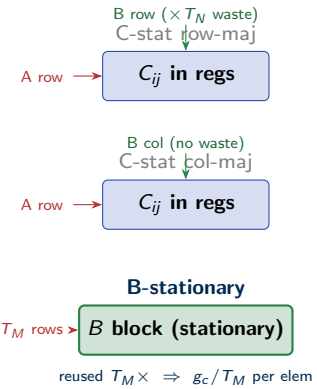
Runtime = whichever is slower

Which Loop Order Fits the FIFO?

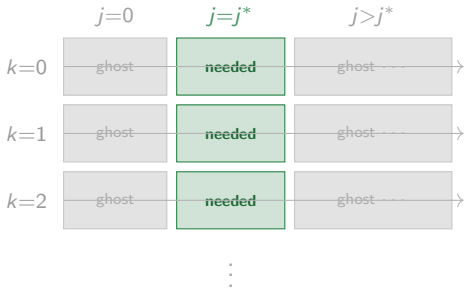
Problem: B elements emerge from the PRNG in a **fixed order** — you cannot skip or replay elements.

Three loop structures are possible:

Mode	In regs	B cost / elem
C-stat row-maj	C_{ij} tile	$g_c \times T_N$ (waste)
C-stat col-maj	C_{ij} tile	g_c
B-stationary	B_{ij} tile	$\boldsymbol{g_c / T_M}$



C-Stationary Row-Major: Ghost Read Problem

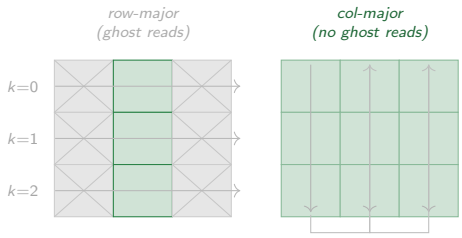


N/T_N groups generated per k step; only 1 is useful

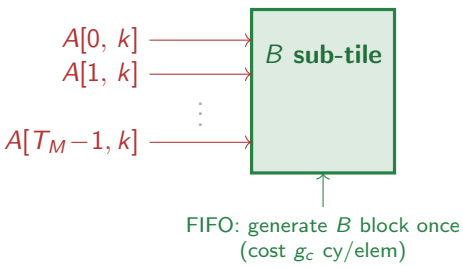
C-Stationary Col-Major: Ghost Reads Gone, Problem Remains

Fix: generate B in col-major order.

The FIFO output now matches C-stationary's consumption order → No ghost reads.



B-Stationary: T_M-Fold Register Reuse



Loop order: B outer, A inner. Each B block is fetched *once*. → Each B element is used T_M times — generation cost amortized:

	g_c=0	g_c=10	g_c=100
B-stat	ref	ref	**ref**
C col-maj	2× fstr	1.28× fstr	7.5× slwr
C row-maj	2× fstr	1.25× fstr	7.5× slwr

Naive Cycle Model: Two Cost Terms

Same reuse argument as traffic, but in **cycles**:

Naively assuming the two costs **add**:

$$\frac{T}{MNK} = \frac{C_A}{T_N} + \frac{g_c}{T_M}$$

FIFO is Async: Two Costs in Parallel

The FIFO generates B **in the background** while the core loads A from L1.

The two costs **overlap** — runtime is determined by whichever finishes last:

$$\frac{T}{MNK} = \max\left\{ \frac{C_A}{T_N},\ \frac{g_c}{T_M} \right\}$$

Two regimes:

$g_c/T_M < C_A/T_N$: A-load bound
$g_c/T_M > C_A/T_N$: B-gen bound

Replacing C_A/T_N with Measured $\alpha(T_M, T_N)$

The C_A/T_N term assumes:

C_A = fixed cost per A element from L1
T_N-fold reuse divides it out cleanly

Problem: actual cycles/MAC depends on tile shape through cache residency, line utilization, and access patterns — not a simple constant divided by T_N.

Solution:

$$\frac{T}{MNK} = \max\left\{\alpha(T_M, T_N),\ \frac{g_c}{T_M}\right\}$$

Isolating α: Set $g_c = 0$

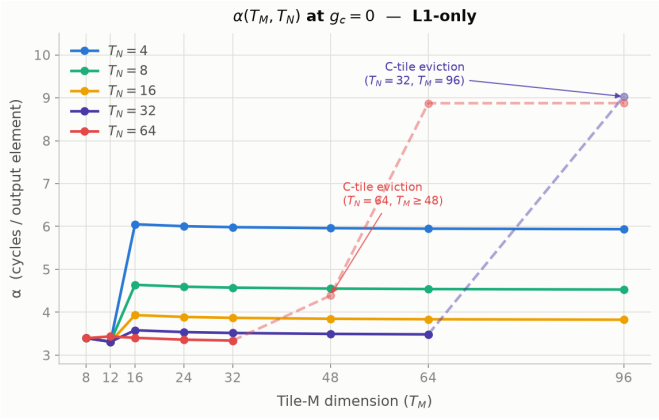
Key trick: set $g_c = 0$

The model collapses:

$$\frac{T}{MNK} = \max\left\{\alpha(T_M, T_N),\ \frac{0}{T_M}\right\} = \alpha(T_M, T_N)$$

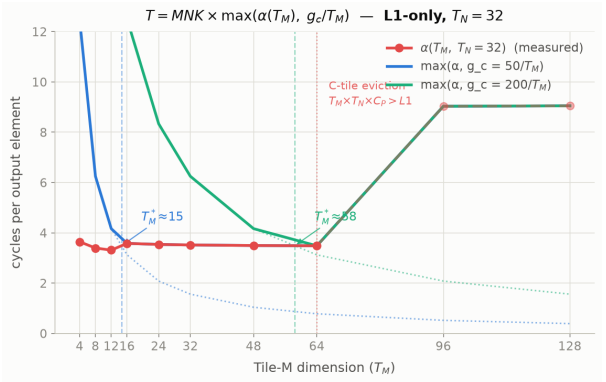
Calibration: Building the α Table

- Small T_N: α spikes when the A tile overflows L1 (T_N=4 at T_M=32)
- Large T_N: α stays low until the C tile overflows L1 (T_N=32 at T_M=96)
- α decreases with T_N (more column reuse)



Using the α Table: Predicting the Optimal Tile

The optimal tile is where the two costs are balanced: $\alpha(T_M^*, T_N^*) \approx g_c / T_M^*$.



Roofline Validation: Setup

Question:

calibrate α once at $g_c = 0$; does the model predict the best tile correctly at any new g_c?

Test conditions:

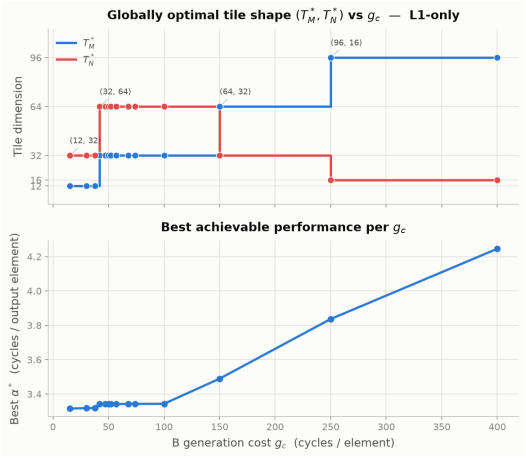
SRAM budgets	64 KB, 128 KB
L1/FIFO splits	4 (64 KB) + 8 (128 KB) = 12 configs
g_c values	10, 100, 250, 280, 300, 310, 325, 350, 500
Total	$12 \times 9 = 108$ conditions

Roofline Validation: Results

	All 108 conditions
T_M^* correct	**108 / 108** (100%)
(T_M^*, T_N^*) exact	**108 / 108** (100%)
Max perf. gap	0%

100% exact match across all hardware configurations and g_c values.

Impact of Tile Selection: Optimal vs. Square



T_N^* (red) drops 64→32→16 as g_c grows.

Impact of Tile Selection: Optimal vs. Square

Question:

Does choosing the right (T_M, T_N) actually matter?

Comparing:

- Optimal asymmetric tile (T_M^*, T_N^*)
- Square tile $(T_M = T_N$, fixed$)$

Speedup from correct tile:

- $g_c = 100$: **20–60%** faster
- $g_c = 250$: $\sim$**85%** faster
- $g_c = 400$: **up to 90%** faster

Gain grows with g_c.

Impact of Tile Selection: Optimal vs. Square